

ORIGINAL RESEARCH

Low-Normal Hemoglobin Concentrations and Clinical Outcomes in Heart Failure

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ABSTRACT

BACKGROUND The World Health Organization (WHO) defines anemia as hemoglobin concentration <12 g/dL in women and <13 g/dL in men. Although widely used clinically, these thresholds were based on distributions in healthy populations rather than on outcomes. Whether these WHO thresholds adequately reflect the relationship between hemoglobin concentration and clinical outcomes in patients with heart failure (HF) is uncertain.

OBJECTIVES We sought to characterize sex-specific associations between hemoglobin concentration and clinical outcomes and to identify the hemoglobin concentrations associated with the lowest observed risk in patients with HF across the spectrum of left ventricular ejection fraction.

METHODS Patient-level data were pooled from 6 trials in heart failure with reduced ejection fraction (HFrEF) and 5 trials in heart failure with preserved ejection fraction (HFpEF) or heart failure with mildly reduced ejection fraction (HFmrEF). Associations among baseline hemoglobin and a first HF hospitalization or cardiovascular death, the components of this composite, and all-cause death were examined using Cox proportional hazards models. Sex-specific hemoglobin concentrations associated with the lowest incidence rates were estimated using Poisson regression with restricted cubic splines. Outcomes were also examined by hemoglobin categories defined relative to WHO anemia thresholds.

RESULTS Overall, 25,003 patients with HFrEF (5,581 women, 19,422 men) and 17,210 with HFpEF/HFmrEF (8,763 women, 8,447 men) were included. In HFrEF, median hemoglobin was 13.0 g/dL (Q1-Q3: 12.1-13.9 g/dL) in women and 14.0 g/dL (Q1-Q3: 12.9-15.1 g/dL) in men; corresponding values in HFpEF/HFmrEF were 13.1 g/dL (Q1-Q3: 12.1-14.0 g/dL) and 14.0 g/dL (Q1-Q3: 12.8-15.0 g/dL), respectively. Across HF phenotypes and outcomes, the lowest risk occurred at hemoglobin concentrations of approximately 14 g/dL in women and 15 g/dL in men, above the WHO anemia thresholds. With hemoglobin ≥ 2 g/dL above the WHO anemia thresholds as the reference, WHO-defined anemia was associated with the highest adjusted risk. Excess risk was also observed at 0 to <1 g/dL above the thresholds.

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ABBREVIATIONS AND ACRONYMS

HF = heart failure

HFmrEF = heart failure with mildly reduced ejection fraction

HFpEF = heart failure with preserved ejection fraction

HFrEF = heart failure with reduced ejection fraction

LVEF = left ventricular ejection fraction

NT-proBNP = N-terminal pro-B-type natriuretic peptide

CONCLUSIONS In patients with HF, hemoglobin concentrations associated with the lowest risk of death and HF hospitalization were approximately 14 g/dL in women and 15 g/dL in men. Hemoglobin concentrations above WHO anemia thresholds may still carry prognostic information and, in conjunction with other clinical findings, may prompt consideration of potentially reversible contributors such as iron deficiency. (Candesartan Cilexetil in Heart Failure Assessment of Reduction in Mortality and Morbidity [CHARM-Added], [NCT00634309](#); Candesartan in Heart Failure Assessment of Reduction in Mortality and Morbidity [CHARM-Alternative], [NCT00634400](#); Candesartan Cilexetil in Heart Failure Assessment of Reduction in Mortality and Morbidity [CHARM-Preserved], [NCT00634712](#); A Comparison Of Outcomes In Patients In New York Heart Association [NYHA] Class II Heart Failure When Treated With Eplerenone Or Placebo In Addition To Standard Heart Failure Medicines [EMPHASIS-HF], [NCT00232180](#); Irbesartan in Heart Failure With Preserved Systolic Function [I-Preserve], [NCT00095238](#); Aldosterone Antagonist Therapy for Adults With Heart Failure and Preserved Systolic Function [TOPCAT], [NCT00094302](#); This Study Will Evaluate the Efficacy and Safety of LCZ696 Compared to Enalapril on Morbidity and Mortality of Patients With Chronic Heart Failure [PARADIGM-HF], [NCT01035255](#); Efficacy and Safety of LCZ696 Compared to Valsartan, on Morbidity and Mortality in Heart Failure Patients With Preserved Ejection Fraction [PARAGON-HF], [NCT01920711](#); Study to Evaluate the Effect of Dapagliflozin on the Incidence of Worsening Heart Failure or Cardiovascular Death in Patients With Chronic Heart Failure [DAPA-HF], [NCT03036124](#); Registrational Study With Omecamtiv Mecarbil [AMG 423] to Treat Chronic Heart Failure With Reduced Ejection Fraction [GALACTIC-HF], [NCT02929329](#); Study to Evaluate the Efficacy [Effect on Disease] and Safety of Finerenone in Participants With Heart Failure and Left Ventricular Ejection Fraction [Proportion of Blood Expelled Per Heart Stroke] Greater or Equal to 40% [FINEARTS-HF], [NCT04435626](#)). (JACC. 2026;■:■-■) © 2026 by the American College of Cardiology Foundation.

Anemia is usually defined using the World Health Organization (WHO) thresholds of hemoglobin <13.0 g/dL (130 g/L) in men aged 15 to 65 years and <12.0 g/dL in nonpregnant women aged 15 to 65 years.¹ These thresholds were defined statistically, using the lower fifth percentile of hemoglobin concentration in the healthy reference populations, although it was recognized that a higher hemoglobin cutoff (ie, 13.5 g/dL) might be appropriate in adult men and that evidence was lacking for older individuals of either sex. The 2024 updated WHO guidance stressed that research is needed on the hemoglobin concentrations below which clinical symptoms and increased risk of morbidity and mortality are observed to better define functional thresholds for anemia.¹ This outcomes-based, rather than statistical, approach may be especially relevant to heart failure (HF), where impaired oxygen delivery represents one plausible mechanism through which lower hemoglobin concentrations may contribute to worse symptoms, reduced functional capacity, and adverse clinical outcomes.²⁻⁵ Previous studies have shown that hemoglobin concentrations above the

conventional WHO anemia thresholds may remain clinically and prognostically relevant in patients referred to specialist clinics with suspected HF and in broader populations with cardiovascular disease.^{6,7} To confirm and extend these observations, we examined the sex-specific, continuous associations between hemoglobin concentration and clinically relevant outcomes, including HF hospitalization and death, using individual participant-level data from multiple randomized controlled trials of patients with established HF across the full spectrum of left ventricular ejection fraction (LVEF).

METHODS

Individual participant-level data were pooled from 6 randomized trials in heart failure with reduced ejection fraction (HFrEF): CHARM (Candesartan in Heart Failure Assessment of Reduction in Mortality and Morbidity)-Added and CHARM-Alternative, EMPHASIS-HF (Eplerenone in Mild Patients Hospitalization and Survival Study in Heart Failure),

The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

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PARADIGM-HF (Prospective Comparison of ARNI With ACEI to Determine Impact on Global Mortality and Morbidity in Heart Failure), DAPA-HF (Dapagliflozin and Prevention of Adverse Outcomes in Heart Failure), and GALACTIC-HF (Global Approach to Lowering Adverse Cardiac Outcomes Through Improving Contractility in Heart Failure). Data were also pooled from 5 trials in heart failure with preserved ejection fraction (HFpEF) or heart failure with mildly reduced ejection fraction (HFmrEF): CHARM-Preserved, I-Preserve (Irbesartan in Heart Failure with Preserved Systolic Function), TOPCAT (Treatment of Preserved Cardiac Function Heart Failure With an Aldosterone Antagonist; Americas only), PARAGON-HF (Prospective Comparison of Angiotensin Receptor Neprilysin Inhibitor With Angiotensin Receptor Blocker Global Outcomes in HF With Preserved Ejection Fraction), and FINEARTS-HF (Finerenone Trial to Investigate Efficacy and Safety Superior to Placebo in Patients with Heart Failure).

The design, baseline characteristics, and primary results of these trials have been published previously.⁸⁻¹⁶ Given the substantial regional heterogeneity in the conduct and results of TOPCAT, the present analysis was restricted to participants enrolled in the Americas. Each trial was approved by the relevant ethics committees, and all participants provided written informed consent.

Eligibility criteria related to hemoglobin concentrations are summarized in [Supplemental Tables 1 and 2](#). I-Preserve excluded patients with substantial laboratory abnormalities, including hemoglobin <11 g/dL, whereas PARAGON-HF and FINEARTS-HF excluded patients with hemoglobin <10 g/dL. For CHARM-Preserved, CHARM-Added, CHARM-Alternative, TOPCAT, DAPA-HF, EMPHASIS-HF, PARADIGM-HF, and GALACTIC-HF, no explicit lower hemoglobin threshold was identified in the published trial reports, design papers, or publicly available protocols reviewed; accordingly, these trials were treated as having no prespecified hemoglobin threshold for exclusion in the present analysis.

TRIAL OUTCOMES. The primary outcome of each trial is summarized in [Supplemental Tables 1 and 2](#). For the present analysis, the primary outcome was the composite of first HF hospitalization or cardiovascular death. Secondary outcomes included first HF hospitalization, cardiovascular death, and all-cause death. Analyses of Kansas City Cardiomyopathy Questionnaire Total Symptom Score (KCCQ-TSS) were restricted to trials with available individual participant-level data, including PARADIGM-HF, TOPCAT, PARAGON-HF, DAPA-HF,

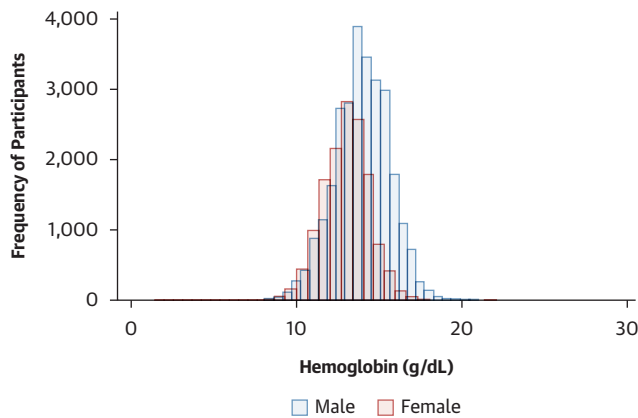
GALACTIC-HF, and FINEARTS-HF. Events and causes of death were adjudicated by central endpoint committees according to prespecified criteria in each trial.

STATISTICAL ANALYSIS. Based on baseline hemoglobin concentrations in each trial, patients were categorized according to hemoglobin concentrations relative to the WHO anemia thresholds as follows: WHO-defined anemia, <12.0 g/dL in women and <13.0 g/dL in men; hemoglobin 0 to <1 g/dL above WHO threshold, 12.0 to <13.0 g/dL in women and 13.0 to <14.0 g/dL in men; hemoglobin 1 to <2 g/dL above WHO threshold, 13.0 to <14.0 g/dL in women and 14.0 to <15.0 g/dL in men; and hemoglobin ≥2 g/dL above WHO threshold, ≥14.0 g/dL in women and ≥15.0 g/dL in men. The number of missing values for each variable across trials is presented in [Supplemental Table 3](#).

Baseline characteristics were summarized as mean ± SD or median (Q1-Q3) for continuous variables and as count and percentage for categorical variables. Trends across ordered hemoglobin categories were assessed using the Jonckheere-Terpstra test for continuous variables and the Cochran-Armitage trend test for binary variables.

For time-to-first-event analyses, Cox proportional hazards models were used to estimate HR and 95% CI according to hemoglobin category. All models were stratified by trial, allowing the baseline hazard to vary across the component trials. The unadjusted models included hemoglobin category and randomized treatment assignment. The multivariable-adjusted models additionally included age, sex, race, heart rate, NYHA functional class III or IV, body mass index, LVEF, serum creatinine, log-transformed N-terminal pro-B-type natriuretic peptide (NT-proBNP), history of atrial fibrillation, history of myocardial infarction, history of diabetes, anticoagulant use, diuretic use, previous HF hospitalization, and history of stroke. Sex was included as a covariate in analyses of the overall population but was omitted from models conducted separately in women and men. Missing NT-proBNP values were handled using the missing-indicator method. Incidence rates were estimated per 100 person-years.

In addition, hemoglobin was modeled as a continuous variable using restricted cubic splines to evaluate the association between hemoglobin concentration and the hazard of each clinical outcome. Both unadjusted and multivariable-adjusted Cox proportional hazards models were fitted. Multivariable models were stratified by trial and adjusted for the covariates described in the previous text. For graphical presentation of the sex-specific analyses, hemoglobin

FIGURE 1 Distribution of Baseline Hemoglobin Concentrations in Patients With Heart Failure Stratified by Sex

concentration was expressed as the difference from the sex-specific WHO anemia threshold (12.0 g/dL in women and 13.0 g/dL in men). In the HR plots, estimates were re-referenced to the corresponding sex-specific WHO threshold, which was assigned an HR of 1.0. The number of knots was selected by comparing candidate spline models with different numbers of knots and choosing the model with the lowest Akaike information criterion; knots were placed at standard quantiles of the hemoglobin distribution according to the selected number of knots.

We further evaluated sex-specific associations of hemoglobin concentration with clinical outcome incidence rates and baseline KCCQ-TSS. Sex-specific unadjusted Poisson regression models incorporating restricted cubic splines for hemoglobin were used to estimate incidence rates per 100 person-years across the spectrum of hemoglobin concentrations, with the log of person-years included as an offset. For graphical presentation, incidence rates were plotted against the baseline hemoglobin concentration. Baseline KCCQ-TSS was analyzed separately as a continuous outcome using sex-specific linear regression models incorporating restricted cubic splines for hemoglobin concentration. The number and placement of knots were determined using the Akaike information criterion-based approach described in the previous text.

To identify the hemoglobin concentration associated with the lowest incidence rate of each clinical outcome, model-derived incidence rates from the unadjusted Poisson models were calculated at 0.01-g/dL increments within the sex-specific 2.5th to 97.5th percentile range of observed hemoglobin

concentrations. The hemoglobin concentration corresponding to the lowest predicted incidence rate was defined as the incidence-rate nadir. To characterize the width of the low-risk region surrounding the nadir, we also performed an exploratory analysis in which we identified the contiguous ranges of hemoglobin concentrations over which the model-predicted incidence rate remained no more than 5% or 10% above the minimum predicted incidence rate (ie, ≤ 1.05 or ≤ 1.10 times the model-derived minimum, respectively). The lower and upper bounds of these ranges were determined using the same 0.01-g/dL increments. The 95% CI for the nadir and the lower and upper bounds of the low-risk ranges were estimated using percentile bootstrap resampling with 500 replications, using the 2.5th and 97.5th percentiles of the bootstrap distribution. As a sensitivity analysis, we additionally fitted sex-specific multivariable Poisson regression models adjusted for the clinical covariates described in the previous text. Population-average incidence rates across baseline hemoglobin concentrations were estimated using marginal standardization over the observed covariate distribution. The corresponding hemoglobin nadirs and 5% and 10% low-risk ranges, together with their 95% CIs, were then rederived from the standardized incidence-rate curves using the same procedures as in the primary analysis. Spline knots were fixed at the values estimated from the original analytic sample. All statistical analyses were performed using Stata version 18.0 (StataCorp LLC), and a 2-sided P value < 0.05 was considered statistically significant.

RESULTS

Among 42,213 patients with HF and available baseline hemoglobin data, 14,344 were women and 27,869 were men; 25,003 had HFrEF and 17,210 had HFpEF/HFmrEF. Overall, 10,559 patients (25.0%) met WHO criteria for anemia. An additional 9,701 participants (23.0%) had hemoglobin concentrations 0 to < 1 g/dL above the WHO anemia thresholds, 10,615 (25.1%) had concentrations 1 to < 2 g/dL above the thresholds, and 11,338 (26.9%) had concentrations ≥ 2 g/dL above the thresholds. Mean hemoglobin concentrations were 13.0 ± 1.5 g/dL (median: 13.0 g/dL; Q1-Q3: 12.1-13.9 g/dL) in women and 14.0 ± 1.7 g/dL (median: 14.0 g/dL; Q1-Q3: 12.9-15.1 g/dL) in men with HFrEF, and 13.0 ± 1.5 g/dL (median: 13.1 g/dL; Q1-Q3: 12.1-14.0 g/dL) and 13.9 ± 1.7 g/dL (median: 14.0 g/dL; Q1-Q3: 12.8-15.0 g/dL), respectively, in those with HFpEF/HFmrEF (Figure 1).

BASELINE CHARACTERISTICS BY HEMOGLOBIN CATEGORY. Baseline characteristics differed across hemoglobin categories (Supplemental Table 4). Patients meeting the WHO criteria for anemia were older and had lower body mass index, lower estimated glomerular filtration rate, and higher NT-proBNP than those with hemoglobin concentrations ≥ 2 g/dL above the thresholds. They also had a higher prevalence of diabetes mellitus, prior myocardial infarction, and previous HF hospitalization, and lower KCCQ-TSS. These patterns were broadly consistent when women and men were examined separately (Table 1). In DAPA-HF, the only trial with available iron indexes, iron deficiency, defined as transferrin saturation $<20\%$, was present in 50%, 34%, 26%, and 22% of patients across progressively higher hemoglobin categories, respectively (Supplemental Table 5).

INCIDENCE RATES ACCORDING TO BASELINE HEMOGLOBIN. When hemoglobin was analyzed as a continuous variable, the incidence rates of first HF hospitalization or cardiovascular death, first HF hospitalization, cardiovascular death, and all-cause death were highest at lower hemoglobin concentrations in both women and men. Incidence rates decreased as hemoglobin increased from 10 g/dL to approximately 14 to 15 g/dL, beyond which they generally plateaued or increased slightly (Figure 2). Similar patterns were observed in patients with HFpEF/HFmrEF and HFrEF (Supplemental Figures 1 and 2).

In the 6 trials with available KCCQ-TSS data, KCCQ-TSS scores increased, indicating better symptom status, as hemoglobin increased from 10 g/dL to approximately 15 g/dL, beyond which they plateaued (Supplemental Figure 3).

HEMOGLOBIN CONCENTRATIONS ASSOCIATED WITH THE LOWEST INCIDENCE RATES AND LOW-RISK RANGES. The hemoglobin concentrations associated with the lowest risk of clinical outcomes were consistently higher than the current WHO anemia thresholds (Table 2, Figures 2 and 3). Among women, the incidence rate nadirs were 14.3 g/dL for first HF hospitalization or cardiovascular death, 14.4 g/dL for first HF hospitalization, 14.1 g/dL for cardiovascular death, and 14.2 g/dL for all-cause death. Among men, the corresponding nadirs were 15.2, 15.2, 15.3, and 15.3 g/dL, respectively. In an exploratory analysis, the model-predicted incidence rate for the primary composite outcome remained within 5% of its minimum across hemoglobin ranges of

13.7 to 15.1 g/dL in women and 14.6 to 15.9 g/dL in men; the corresponding 10% ranges were 13.5 to 15.5 g/dL and 14.4 to 16.3 g/dL, respectively. Similar ranges were observed for the secondary outcomes (Supplemental Table 6), indicating a broad low-risk region around each nadir. These nadirs were broadly consistent across HF phenotypes, with values of approximately 14 g/dL in women and 15 g/dL in men in both HFpEF/HFmrEF and HFrEF (Supplemental Tables 7 and 8). In adjusted sensitivity analyses using marginal standardization, the estimated nadirs were generally similar to those in the primary unadjusted analyses, except for cardiovascular death in women, for which the adjusted nadir was lower and less precisely estimated (Supplemental Table 9).

CLINICAL OUTCOMES ACCORDING TO HEMOGLOBIN CATEGORY. For the primary outcome of first HF hospitalization or cardiovascular death, incidence rates were 18.8 per 100 person-years (95% CI: 18.2-19.4 per 100 person-years), 12.0 per 100 person-years (95% CI: 11.6-12.5 per 100 person-years), 9.8 per 100 person-years (95% CI: 9.4-10.2 per 100 person-years), and 9.8 per 100 person-years (95% CI: 9.4-10.2 per 100 person-years) among patients meeting the WHO criteria for anemia, those with hemoglobin concentrations 0 to <1 g/dL above the thresholds, those 1 to <2 g/dL above the thresholds, and those ≥ 2 g/dL above the thresholds, respectively. Compared with patients with hemoglobin concentrations ≥ 2 g/dL above the thresholds, those meeting WHO criteria for anemia had a higher risk of the primary outcome after multivariable adjustment (adjusted HR: 1.35; 95% CI: 1.28-1.43). Patients with hemoglobin concentrations 0 to <1 g/dL above the thresholds also had a higher adjusted risk (adjusted HR: 1.12; 95% CI: 1.06-1.18), whereas those 1 to <2 g/dL above the thresholds had a risk similar to the reference group (Supplemental Table 10).

Similar patterns were observed for first HF hospitalization, cardiovascular death, and all-cause death. Compared with patients with hemoglobin concentrations ≥ 2 g/dL above the thresholds, those with hemoglobin concentrations 0 to <1 g/dL above the thresholds had higher adjusted risks of first HF hospitalization, cardiovascular death, and all-cause death. In contrast, patients with hemoglobin concentrations 1 to <2 g/dL above the thresholds generally had risks similar to those of the reference group (Supplemental Table 10). In a sensitivity

TABLE 1 Baseline Characteristics by Hemoglobin Category in Women and Men

	Women			
	WHO-Defined Anemia ^a (n = 3,172)	Hb 0- $<$ 1 g/dL Above WHO Threshold ^a (n = 3,626)	Hb 1- $<$ 2 g/dL Above WHO Threshold ^a (n = 3,956)	Hb $\geq$ 2 g/dL Above WHO Threshold ^a (n = 3,590)
Age, y	72 (65-79)	72 (64-78)	71 (64-77)	71 (64-77)
Race				
White	2,202 (69)	2,783 (77)	3,210 (81)	3,019 (84)
Black	330 (10)	242 (7)	211 (5)	129 (4)
Asian	478 (15)	390 (11)	320 (8)	207 (6)
Other race	162 (5)	211 (6)	215 (5)	235 (7)
Heart failure characteristics				
NYHA functional class III/IV	1,348 (43)	1,479 (41)	1,700 (43)	1,721 (48)
LVEF, %	50 (32-60)	50 (31-60)	50 (32-60)	50 (33-60)
KCCQ total symptom score	68 (47-85)	69 (50-85)	72 (53-88)	71 (51-86)
Physiological and laboratory measurements				
Systolic blood pressure, mm Hg	129 (116-140)	130 (118-140)	130 (118-140)	130 (120-140)
Heart rate, beats/min	70 (64-79)	70 (63-78)	70 (64-79)	72 (64-80)
Body mass index, kg/m ²	29 (25-34)	29 (25-34)	29 (26-34)	30 (26-34)
Body mass index groups, kg/m ²				
$<$ 18.5 (underweight)	70 (2)	75 (2)	49 (1)	47 (1)
18.5- $<$ 25 (normal weight)	795 (25)	811 (22)	797 (20)	673 (19)
25- $<$ 30 (overweight)	954 (30)	1,127 (31)	1,299 (33)	1,154 (32)
$\geq$ 30 (obesity)	1,344 (42)	1,603 (44)	1,803 (46)	1,711 (48)
eGFR, ^b mL/min/1.73 m ²	54 (41-70)	60 (47-76)	65 (51-78)	65 (51-78)
NT-proBNP, pg/mL	1,415 (642-2,967)	1,084 (496-2,098)	963 (404-1,923)	1,020 (412-2,025)
Potassium, mmol/L	4.4 (4.1-4.7)	4.4 (4.1-4.7)	4.4 (4.1-4.7)	4.4 (4.1-4.7)
Creatinine, μ mol/L	91 (72-115)	82 (67-102)	80 (66-96)	80 (66-95)
HbA1c, %	6.3 (5.8-7.1)	6.1 (5.7-6.9)	6.1 (5.7-6.8)	6.1 (5.8-6.8)
Medical history				
Hypertension	2,666 (84)	3,015 (83)	3,299 (83)	3,067 (85)
Diabetes mellitus	1,501 (47)	1,335 (37)	1,308 (33)	1,079 (30)
Myocardial infarction	849 (27)	891 (25)	941 (24)	758 (21)
AF (history)	1,211 (38)	1,383 (38)	1,508 (38)	1,613 (45)
Any prior hospitalization for HF	2,009 (63)	2,108 (58)	2,224 (56)	1,990 (55)
Chronic obstructive pulmonary disease ^c	307 (10)	365 (11)	355 (10)	351 (10)
Stroke ^d	346 (11)	358 (10)	384 (10)	350 (10)
Treatments				
Beta-blocker	2,561 (81)	2,888 (80)	3,173 (80)	2,841 (79)
ACEI, ARB, or ARNI ^e	2,355 (88)	2,619 (90)	2,680 (90)	2,279 (91)
Mineralocorticoid antagonist ^f	851 (42)	1,063 (41)	1,147 (39)	1,083 (40)
Anticoagulant agent	1,165 (38)	1,260 (36)	1,371 (36)	1,357 (39)
Antiplatelet agent	1,264 (40)	1,421 (39)	1,484 (38)	1,305 (36)
Digitalis	513 (16)	599 (17)	652 (16)	775 (22)
Diuretic agents	2,917 (92)	3,287 (91)	3,520 (89)	3,209 (89)
ICD or CRT-D	228 (7)	222 (6)	232 (6)	196 (5)

Values are median (Q1-Q3) or n (%). ^aCategories were defined as follows: World Health Organization (WHO)-defined anemia, $<$ 12.0 g/dL in women and $<$ 13.0 g/dL in men; hemoglobin (Hb) 0- $<$ 1 g/dL above the WHO threshold, 12.0- $<$ 13.0 g/dL in women and 13.0- $<$ 14.0 g/dL in men; Hb 1- $<$ 2 g/dL above the WHO threshold, 13.0- $<$ 14.0 g/dL in women and 14.0- $<$ 15.0 g/dL in men; and Hb $\geq$ 2 g/dL above the WHO threshold, $\geq$ 14.0 g/dL in women and $\geq$ 15.0 g/dL in men. ^bIn CHARM (Candesartan in Heart Failure Assessment of Reduction in Mortality and Morbidity), the estimated glomerular filtration rate (eGFR) was calculated using the 2021 Chronic Kidney Disease Epidemiology Collaboration creatinine equation. ^cIn I-Preserve (Irbesartan in Heart Failure with Preserved Systolic Function), asthma was included in the definition of chronic obstructive pulmonary disease. ^dIn I-Preserve, transient ischemic attack was included in the definition of stroke. ^eParticipants in CHARM and I-Preserve were excluded from this analysis. In PARAGON-HF (Prospective Comparison of Angiotensin Receptor Neprilysin Inhibitor With Angiotensin Receptor Blocker Global Outcomes in HF With Preserved Ejection Fraction) and PARADIGM-HF (Prospective Comparison of ARNI With ACEI to Determine Impact on Global Mortality and Morbidity in Heart Failure), use of angiotensin-converting enzyme inhibitor (ACEI), angiotensin receptor blocker (ARB), or angiotensin receptor-neprilysin inhibitor (ARNI) was assigned according to randomized treatment allocation. ^fParticipants in TOPCAT (Treatment of Preserved Cardiac Function Heart Failure With an Aldosterone Antagonist; Americas only), EMPHASIS-HF (Eplerenone in Mild Patients Hospitalization and Survival Study in Heart Failure), and FINEARTS-HF (Finerenone Trial to Investigate Efficacy and Safety Superior to Placebo in Patients with Heart Failure) were excluded. Only spironolactone was regarded as a mineralocorticoid antagonist in I-Preserve and CHARM.

AF = atrial fibrillation; HbA1c = glycated hemoglobin; KCCQ = Kansas City Cardiomyopathy Questionnaire; LVEF = left ventricular ejection fraction; NT-proBNP = N-terminal pro-B-type natriuretic peptide.

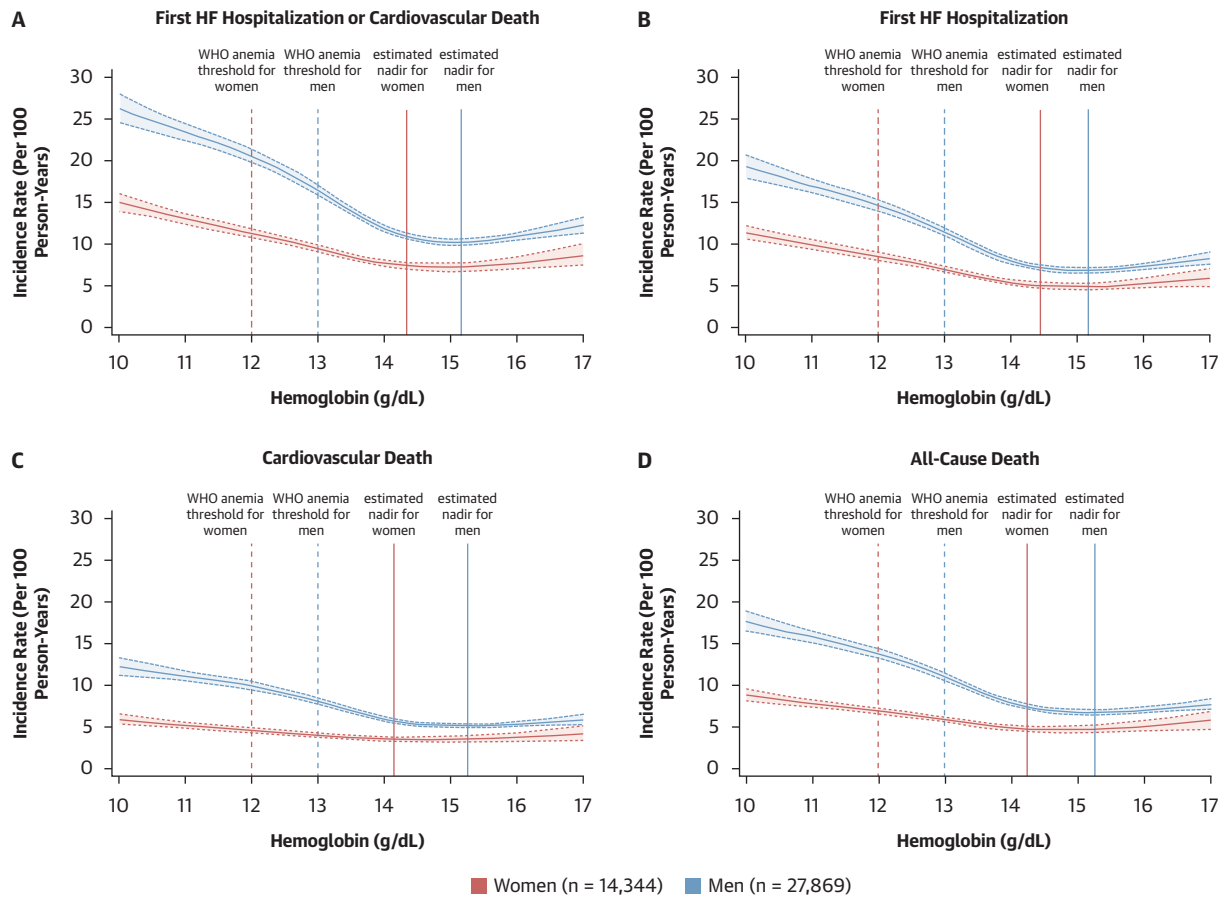
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TABLE 1 Continued

	Men			
	WHO-Defined Anemia ^a (n = 7,387)	Hb 0-<1 g/dL Above WHO Threshold ^a (n = 6,075)	Hb 1-<2 g/dL Above WHO Threshold ^a (n = 6,659)	Hb ≥2 g/dL Above WHO Threshold ^a (n = 7,748)
Age, y	71 (63-77)	68 (61-75)	67 (59-74)	64 (57-71)
Race				
White	5,420 (73)	4,672 (77)	5,235 (79)	6,140 (79)
Black	436 (6)	280 (5)	233 (3)	211 (3)
Asian	1,211 (16)	841 (14)	887 (13)	935 (12)
Other race	320 (4)	282 (5)	304 (5)	462 (6)
Heart failure characteristics				
NYHA functional class III/IV	2,679 (36)	1,961 (32)	2,111 (32)	2,748 (35)
LVEF, %	33 (26-49)	33 (25-45)	33 (26-45)	32 (25-45)
KCCQ total symptom score	75 (54-90)	79 (60-94)	81 (62-94)	79 (60-94)
Physiological and laboratory measurements				
Systolic blood pressure, mm Hg	121 (110-134)	123 (110-135)	123 (111-135)	124 (111-135)
Heart rate, beats/min	70 (62-78)	70 (62-78)	70 (62-78)	71 (64-80)
Body mass index, kg/m ²	27 (24-31)	28 (25-31)	28 (25-32)	29 (26-32)
Body mass index groups, kg/m ²				
<18.5 (underweight)	143 (2)	71 (1)	45 (1)	36 (0)
18.5-<25 (normal weight)	2,278 (31)	1,614 (27)	1,525 (23)	1,436 (19)
25-<30 (overweight)	2,685 (36)	2,311 (38)	2,615 (39)	3,118 (40)
≥30 (obesity)	2,261 (31)	2,070 (34)	2,458 (37)	3,144 (41)
eGFR, ^b mL/min/1.73 m ²	58 (44-74)	66 (52-80)	68 (54-83)	70 (56-83)
NT-proBNP, pg/mL	1,880 (913-3,943)	1,394 (742-2,716)	1,263 (667-2,428)	1,211 (636-2,262)
Potassium, mmol/L	4.5 (4.1-4.8)	4.5 (4.2-4.8)	4.4 (4.2-4.8)	4.5 (4.2-4.8)
Creatinine, μmol/L	109 (88-139)	99 (81-120)	96 (80-115)	96 (80-115)
HbA1c, %	6.3 (5.8-7.2)	6.2 (5.8-6.9)	6.1 (5.7-6.8)	6.1 (5.7-6.9)
Medical history				
Hypertension	5,634 (76)	4,528 (75)	4,990 (75)	5,830 (75)
Diabetes mellitus	3,633 (49)	2,377 (39)	2,330 (35)	2,513 (32)
Myocardial infarction	3,335 (45)	2,732 (45)	2,826 (42)	3,119 (40)
AF (history)	3,207 (43)	2,449 (40)	2,640 (40)	3,296 (43)
Any prior hospitalization for HF	5,125 (69)	3,902 (64)	4,241 (64)	5,008 (65)
Chronic obstructive pulmonary disease ^c	1,179 (17)	865 (15)	900 (15)	1,029 (14)
Stroke ^d	829 (11)	611 (10)	650 (10)	736 (10)
Treatments				
Beta-blocker	6,349 (86)	5,331 (88)	5,835 (88)	6,691 (86)
ACEI, ARB, or ARNI ^e	5,954 (89)	4,986 (93)	5,433 (94)	6,226 (94)
Mineralocorticoid antagonist ^f	3,060 (55)	2,672 (57)	2,923 (56)	3,508 (56)
Anticoagulant agent	3,439 (50)	2,690 (48)	2,859 (47)	3,477 (48)
Antiplatelet agent	3,785 (51)	3,058 (50)	3,330 (50)	3,673 (47)
Digitalis	1,399 (19)	1,290 (21)	1,358 (20)	1,870 (24)
Diuretic agents	6,762 (92)	5,391 (89)	5,778 (87)	6,696 (86)
ICD or CRT-D	1,350 (18)	1,016 (17)	1,118 (17)	1,151 (15)

analysis separating participants with hemoglobin concentrations above the sex-specific upper limits of normal (≥15.1 g/dL in women and ≥17.2 g/dL in men), the associations observed in the lower hemoglobin

categories were materially unchanged. Participants with hemoglobin concentrations above the upper limits of normal had a higher adjusted risk of cardiovascular death and directionally higher risks of

FIGURE 2 Incidence Rates of Clinical Outcomes Across the Spectrum of Hemoglobin Concentrations in Patients With HF Stratified by Sex

The primary composite outcome (A), first heart failure (HF) hospitalization (B), cardiovascular death (C), and all-cause death (D). Sex-specific incidence rates according to baseline hemoglobin concentration. Baseline hemoglobin concentration (g/dL) is shown on the x-axis, and the incidence rate for each outcome is shown on the y-axis. Red solid vertical lines indicate the estimated nadir for women, and blue solid vertical lines indicate the estimated nadir for men. Red dashed vertical lines indicate the World Health Organization (WHO) anemia threshold for women (12.0 g/dL), and blue dashed vertical lines indicate the WHO anemia threshold for men (13.0 g/dL). The solid line shows the estimated incidence rate, and the dashed lines show the 95% CI. All of the incidence rates are shown per 100 person-years.

the other clinical outcomes compared with those with higher hemoglobin concentrations within the normal range ([Supplemental Table 11](#)).

ANALYSES ACCORDING TO HF PHENOTYPE AND SEX. The association between lower hemoglobin categories and higher risk was consistent across HF phenotypes. For the primary outcome, the adjusted HRs among patients meeting WHO criteria for anemia were 1.37 (95% CI: 1.25-1.50) in HFpEF/HFmrEF and 1.31 (95% CI: 1.22-1.41) in HFrEF. Patients with hemoglobin concentrations 0 to <1 g/dL above the thresholds were also at higher adjusted risk in both

HFpEF/HFmrEF and HFrEF, with adjusted HRs of 1.14 (95% CI: 1.04-1.26) and 1.09 (95% CI: 1.01-1.17), respectively ([Supplemental Tables 12 and 13](#)). In both HFpEF/HFmrEF and HFrEF, KCCQ-TSS scores increased, indicating better symptom status, as hemoglobin increased from 10 g/dL to approximately 15 g/dL, beyond which they plateaued or decreased slightly ([Supplemental Figure 3](#)).

When analyses were stratified by sex, lower hemoglobin categories were associated with higher risk in both women and men ([Table 3](#)). For the primary outcome, the adjusted HRs among patients meeting WHO criteria for anemia were 1.33 (95% CI: 1.20-1.48)

in women and 1.36 (95% CI: 1.27-1.45) in men. Patients with hemoglobin concentrations 0 to <1 g/dL above the thresholds were also at higher risk in both women and men. KCCQ-TSS scores increased, indicating better symptom status, as hemoglobin increased from 10 g/dL to approximately 14 g/dL in women and 15 g/dL in men; above these levels, scores plateaued in men and tended to decrease in women (Figure 4).

ANALYSES ACCORDING TO HF SEVERITY AND CONGESTION-RELATED MARKERS. For the primary outcome of first HF hospitalization or cardiovascular death, patients meeting WHO criteria for anemia had higher adjusted risks in both NYHA functional class I/II and III/IV, although the magnitude of association was greater in NYHA functional class III/IV. Excess risk was also observed among patients with hemoglobin concentrations 0 to <1 g/dL above the WHO thresholds, with adjusted HRs of 1.09 (95% CI: 1.00-1.18) and 1.14 (95% CI: 1.05-1.23), respectively. Similar findings were observed when patients were stratified by NT-proBNP and by diuretic use (Supplemental Tables 14 to 16).

DISCUSSION

In this pooled analysis of individual participant data from 42,213 patients enrolled in 11 randomized trials encompassing the full spectrum of LVEF, we found that hemoglobin concentrations associated with the lowest risk of death and HF hospitalization and with the best health status were consistently higher than the current WHO thresholds used to define anemia. Although the model-derived nadirs occurred at approximately 14 g/dL in women and 15 g/dL in men, exploratory analyses identified broad low-risk plateaus around these values. Nevertheless, excess risk remained evident not only among patients meeting the WHO criteria for anemia but also among those with hemoglobin concentrations 0 to <1 g/dL above the WHO thresholds. Our findings are consistent with those of Cleland et al⁷ and Graham et al⁶ and extend their observations by demonstrating the prognostic relevance of borderline or low-normal hemoglobin concentrations across the full spectrum of LVEF and multiple adjudicated outcomes. These observations suggest that hemoglobin concentrations immediately above the current WHO anemia thresholds may remain prognostically relevant in patients with HF.

The 2024 WHO guidance update stated that its hemoglobin cutoffs were derived from the lower fifth percentile of hemoglobin distributions in reference populations presumed to be healthy, rather than

TABLE 2 Hemoglobin Concentrations Associated With the Lowest Incidence Rates of Clinical Outcomes in Women and Men With Heart Failure (Women: n = 14,344; Men: n = 27,869)

Outcome	Number of Events	Hemoglobin at Nadir, g/dL (95% CI)	Predicted Incidence Rate at Nadir, per 100 Person-Years (95% CI)
First HF hospitalization or CV death			
Women	3,358	14.3 (14.1-14.6)	7.22 (6.77-7.66)
Men	7,726	15.2 (15.1-15.4)	10.05 (9.67-10.43)
First HF hospitalization			
Women	2,458	14.4 (14.1-14.6)	5.00 (4.62-5.38)
Men	5,316	15.2 (15.1-15.4)	6.67 (6.33-7.00)
CV death			
Women	1,565	14.1 (13.9-14.6)	3.35 (3.07-3.64)
Men	4,200	15.3 (15.1-15.6)	4.98 (4.74-5.25)
All-cause death			
Women	2,300	14.2 (14.0-14.5)	4.67 (4.32-4.97)
Men	5,713	15.3 (15.2-15.6)	6.57 (6.27-6.86)

The sex-specific sample sizes represent the analytic populations used to fit the corresponding models. The number of events represents the total number of events in each sex-specific analytic population used to fit the respective outcome model, rather than the number of events occurring at the nadir hemoglobin concentration. Predicted incidence rates were derived from sex-specific Poisson regression models incorporating restricted cubic splines and are expressed as events per 100 person-years.

CV = cardiovascular; HF = heart failure.

from the point at which morbidity or mortality risk begins to rise.¹ The WHO working group called for further research to identify functional, outcome-based thresholds, particularly in groups not well represented in the original reference populations.¹ Our findings directly address this knowledge gap by demonstrating that, in patients with HF, excess risk becomes apparent at hemoglobin concentrations above the conventional WHO thresholds.

We also found consistent evidence of a sex difference in the hemoglobin concentration associated with the lowest risk. This is consistent with known sex differences in baseline hemoglobin concentration, likely reflecting androgen-mediated stimulation of erythropoiesis and differences in lean body mass and oxygen demand between men and women.¹⁷ However, the apparently steeper unadjusted mortality gradient across hemoglobin categories in men was substantially attenuated after adjustment for differences in baseline characteristics, HF severity, comorbidities, and treatment. Moreover, there was no significant interaction between sex and hemoglobin category for any outcome. Thus, although the absolute hemoglobin concentration associated with the lowest observed risk differed between women and men, the overall association between lower hemoglobin concentration and prognosis was broadly similar in both sexes.

TABLE 3 Clinical Outcomes According to Hemoglobin Threshold in Patients With HF Stratified by Sex

	Women			
	WHO-Defined Anemia ^a (n = 3,172)	Hb 0-<1 g/dL Above WHO Threshold ^a (n = 3,626)	Hb 1-<2 g/dL Above WHO Threshold ^a (n = 3,956)	Hb ≥2 g/dL Above WHO Threshold ^a (n = 3,590)
First HF hospitalization or cardiovascular death				
Events	981 (31)	828 (23)	798 (20)	751 (21)
Incidence rate (95% CI)	14.8 (13.9-15.7)	9.6 (9.0-10.3)	7.9 (7.4-8.5)	7.9 (7.3-8.5)
Unadjusted HR (95% CI) ^b	1.70 (1.54-1.87)	1.16 (1.05-1.28)	0.99 (0.90-1.10)	Reference
Adjusted HR (95% CI) ^c	1.33 (1.20-1.48)	1.10 (1.00-1.22)	1.05 (0.95-1.16)	Reference
First HF hospitalization				
Events	763 (24)	603 (17)	569 (14)	523 (15)
Incidence rate (95% CI)	11.5 (10.7-12.4)	7.0 (6.5-7.6)	5.7 (5.2-6.1)	5.5 (5.0-6.0)
Unadjusted HR (95% CI) ^b	1.83 (1.63-2.05)	1.19 (1.06-1.34)	1.01 (0.89-1.13)	Reference
Adjusted HR (95% CI) ^c	1.44 (1.28-1.60)	1.13 (1.00-1.27)	1.09 (0.96-1.23)	Reference
Cardiovascular death				
Events	419 (13)	390 (11)	390 (10)	366 (10)
Incidence rate (95% CI)	5.5 (5.0-6.0)	4.1 (3.7-4.5)	3.6 (3.3-4.0)	3.6 (3.2-3.9)
Unadjusted HR (95% CI) ^b	1.52 (1.31-1.75)	1.15 (1.00-1.33)	1.03 (0.89-1.18)	Reference
Adjusted HR (95% CI) ^c	1.09 (0.94-1.27)	1.07 (0.92-1.24)	1.04 (0.90-1.20)	Reference
All-cause death				
Events	645 (20)	588 (16)	564 (14)	503 (14)
Incidence rate (95% CI)	8.4 (7.8-9.1)	6.2 (5.7-6.7)	5.2 (4.8-5.6)	4.9 (4.5-5.3)
Unadjusted HR (95% CI) ^b	1.64 (1.46-1.85)	1.24 (1.10-1.40)	1.07 (0.94-1.20)	Reference
Adjusted HR (95% CI) ^c	1.20 (1.06-1.37)	1.15 (1.01-1.30)	1.08 (0.95-1.22)	Reference

Values are n (%) unless otherwise indicated. ^aCategories were defined as follows: WHO-defined anemia, <12.0 g/dL in women and <13.0 g/dL in men; Hb 0-<1 g/dL above the WHO threshold, 12.0-<13.0 g/dL in women and 13.0-<14.0 g/dL in men; Hb 1-<2 g/dL above the WHO threshold, 13.0-<14.0 g/dL in women and 14.0-<15.0 g/dL in men; and Hb ≥2 g/dL above the WHO threshold, ≥14.0 g/dL in women and ≥15.0 g/dL in men. ^bModels were stratified by trial and adjusted for treatment arm. ^cModels were stratified by trial and adjusted for age, heart rate, NYHA functional classes III or IV, body mass index, LVEF, serum creatinine, NT-proBNP (log-transformed), history of AF, history of myocardial infarction, history of diabetes, history of stroke, previous HF hospitalization, race, anticoagulant use, diuretic use, and treatment arm. In cases with missing data, the missing indicator method was used for NT-proBNP values. Incidence rate is the number of events per 100 person-years.

Abbreviations as in Table 1.

Continued on the next page

Hemoglobin concentration is determined by both red-cell mass and plasma volume and should therefore be considered one component of a broader, multifactorial clinical phenotype. A lower hemoglobin concentration may reflect reduced red-cell mass, plasma-volume expansion, hemodilution, venous congestion, iron deficiency, inflammation, renal dysfunction, nutritional deficiency, or greater overall disease severity.

The association between hemoglobin concentration and clinical outcomes persisted after adjustment for multiple prognostic factors, including NYHA functional class, NT-proBNP, previous HF hospitalization, renal function, and diuretic use, and was broadly similar in analyses stratified by NYHA functional class, NT-proBNP, and diuretic use. Nevertheless, these variables are indirect measures and do not directly quantify red-cell mass, plasma volume, iron status, venous pressure, or congestion. Residual

confounding, plasma-volume expansion, and congestion-related hemodilution may therefore account for at least part of the observed association.

Impaired oxygen delivery is one possible mechanism linking lower hemoglobin concentrations to adverse outcomes; however, the present analysis cannot distinguish this mechanism from the other potential explanations described above.

At the upper end of the hemoglobin distribution, a modest increase in risk was observed in some analyses. This finding should be interpreted cautiously, as it may also reflect residual confounding related to conditions associated with elevated hemoglobin, including smoking, chronic lung disease, and hemoconcentration.

The clinical implications of our findings require cautious interpretation. Hemoglobin concentrations just above the WHO anemia thresholds may be associated with increased risk, but the observed

TABLE 3 Continued

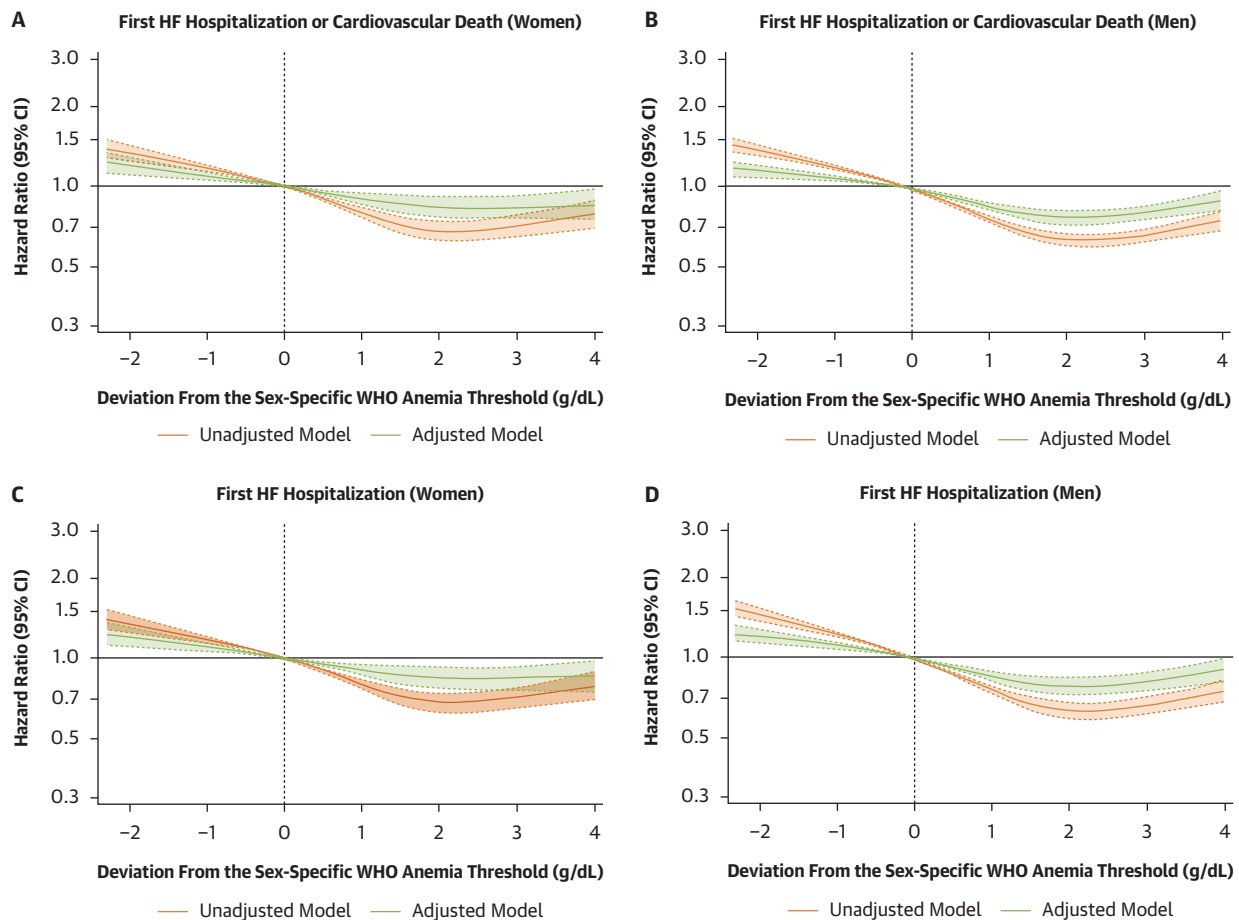
	Men				P Value for Interaction
	WHO-Defined Anemia ^a (n = 7,387)	Hb 0-1 g/dL Above WHO Threshold ^a (n = 6,075)	Hb 1-2 g/dL Above WHO Threshold ^a (n = 6,659)	Hb ≥2 g/dL Above WHO Threshold ^a (n = 7,748)	
First HF hospitalization or cardiovascular death					
Events	2,692 (36)	1,661 (27)	1,554 (23)	1,819 (23)	
Incidence rate (95% CI)	20.8 (20.0-21.6)	13.7 (13.1-14.4)	11.2 (10.6-11.7)	10.9 (10.4-11.4)	
Unadjusted HR (95% CI) ^b	1.85 (1.74-1.96)	1.25 (1.17-1.34)	1.03 (0.96-1.10)	Reference	0.66
Adjusted HR (95% CI) ^c	1.36 (1.27-1.45)	1.12 (1.05-1.21)	1.01 (0.94-1.08)	Reference	0.76
First HF hospitalization					
Events	1,925 (26)	1,117 (18)	1,062 (16)	1,212 (16)	
Incidence rate (95% CI)	14.9 (14.2-15.5)	9.2 (8.7-9.8)	7.6 (7.2-8.1)	7.2 (6.8-7.7)	
Unadjusted HR (95% CI) ^b	1.92 (1.78-2.06)	1.24 (1.15-1.35)	1.05 (0.97-1.14)	Reference	0.93
Adjusted HR (95% CI) ^c	1.38 (1.28-1.50)	1.11 (1.02-1.21)	1.03 (0.95-1.12)	Reference	0.59
Cardiovascular death					
Events	1,506 (20)	903 (15)	826 (12)	965 (13)	
Incidence rate (95% CI)	10.0 (9.5-10.5)	6.8 (6.3-7.2)	5.4 (5.1-5.8)	5.3 (5.0-5.7)	
Unadjusted HR (95% CI) ^b	1.94 (1.79-2.10)	1.29 (1.18-1.42)	1.04 (0.95-1.14)	Reference	0.03
Adjusted HR (95% CI) ^c	1.36 (1.24-1.49)	1.12 (1.01-1.23)	0.99 (0.90-1.10)	Reference	0.34
All-cause death					
Events	2131 (29)	1199 (20)	1095 (16)	1288 (17)	
Incidence rate (95% CI)	14.2 (13.6-14.8)	9.0 (8.5-9.5)	7.2 (6.8-7.7)	7.1 (6.7-7.5)	
Unadjusted HR (95% CI) ^b	2.02 (1.89-2.17)	1.28 (1.19-1.39)	1.03 (0.95-1.12)	Reference	0.008
Adjusted HR (95% CI) ^c	1.42 (1.31-1.53)	1.09 (1.01-1.19)	0.98 (0.90-1.06)	Reference	0.10

associations do not establish a new diagnostic threshold, demonstrate that increasing hemoglobin improves clinical outcomes, or define a treatment target. Nevertheless, low or borderline hemoglobin concentrations may prompt consideration of potentially reversible contributors, including iron deficiency, renal dysfunction, nutritional deficiencies, bleeding, inflammation, and congestion, in the context of the overall clinical assessment. This may be particularly relevant because iron deficiency can occur in patients with HF who do not meet the WHO criteria for anemia, as previously reported by Cleland et al⁷ and Graham et al⁶ and as observed among patients with available iron indices in DAPA-HF.

However, other pharmacological interventions that increase hemoglobin concentration are not generally recommended in HF. Although the erythropoiesis-stimulating agent (ESA) darbepoetin-alfa improved the KCCQ overall summary score and symptom frequency score in the RED-HF (Reduction of Events by Darbepoetin Alfa in Heart Failure) trial, suggesting that anemia does contribute to symptom burden in HFrEF, darbepoetin did not reduce the risk

of death in RED-HF.¹⁸ Although a recent meta-analysis of several trials, including RED-HF, suggested that ESA treatment leads to a significant reduction in HF hospitalization, the increased risk of thromboembolism with these agents precludes their general use in HF.¹⁹ There are no substantial randomized trials with hypoxia-inducible factor prolyl hydroxylase inhibitors reporting clinical outcomes in HF, although a post hoc analysis of a large randomized comparison of the ESA darbepoetin with the hypoxia-inducible factor prolyl hydroxylase inhibitor daprodustat in patients with CKD not on dialysis reported an increase in total (first and recurrent) HF hospitalizations in patients treated with daprodustat.²⁰

STUDY STRENGTHS AND LIMITATIONS. This analysis has several strengths, including its large size; the pooling of individual participant data from trials spanning the full spectrum of LVEF; centrally adjudicated outcomes using prespecified, harmonized definitions; and the use of restricted cubic splines with bootstrapped CIs to identify nadir hemoglobin concentrations without imposing an arbitrary

FIGURE 3 HRs for Clinical Outcomes According to Deviation in Hemoglobin From the Sex-Specific WHO Anemia Threshold

The primary composite outcome in women (A), the primary composite outcome in men (B), first HF hospitalization in women (C), first HF hospitalization in men (D), cardiovascular death in women (E), cardiovascular death in men (F), all-cause death in women (G), and all-cause death in men (H). The solid line shows the spline-estimated HR, and the dashed lines show the 95% CI. The sex-specific WHO anemia threshold (12.0 g/dL in women and 13.0 g/dL in men) was used as the reference value (HR: 1.0). Abbreviations as in [Figure 2](#).

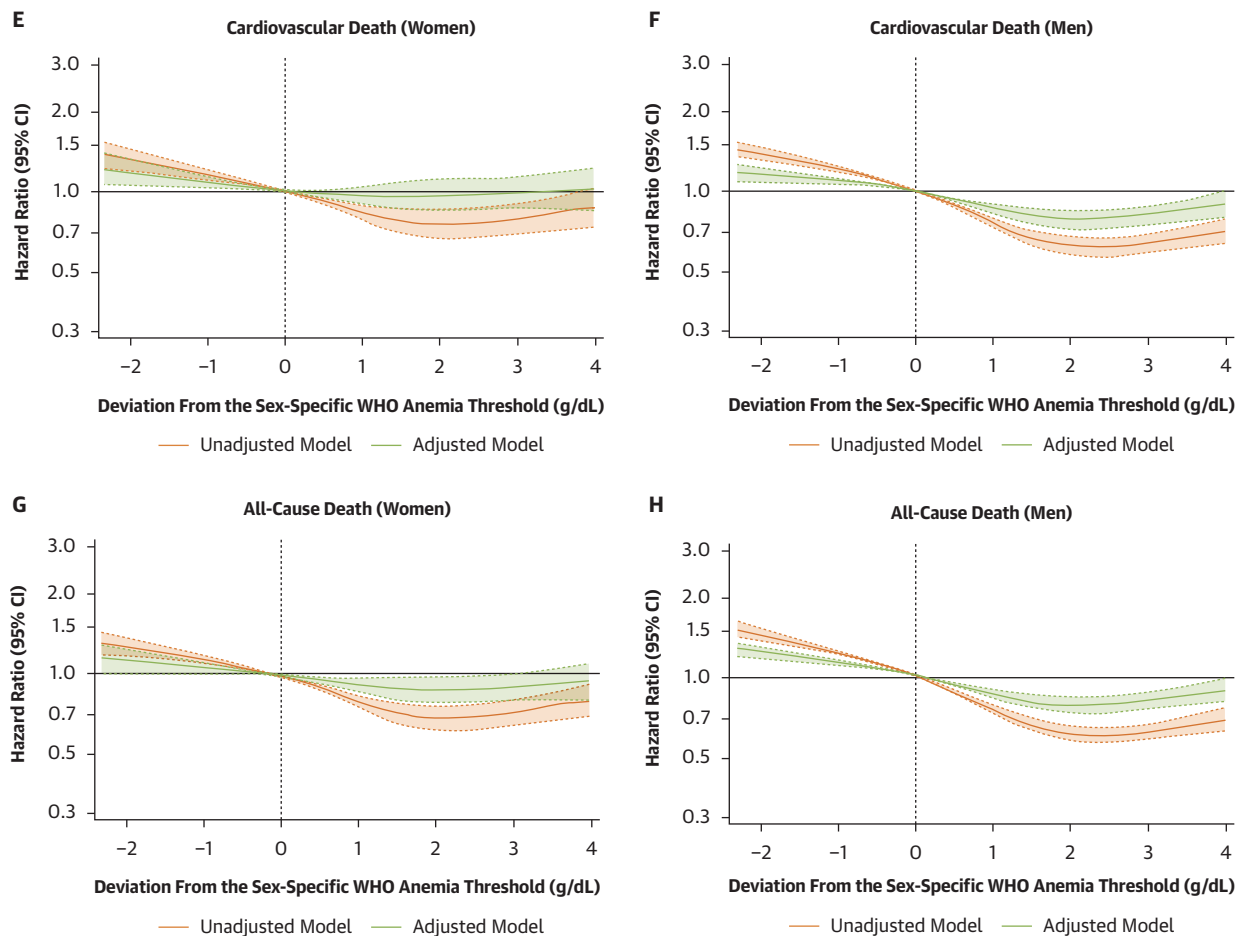
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categorical structure on the data. The consistency of our findings across HF phenotypes and between unadjusted and multivariable-adjusted models strengthens confidence in the robustness of the association.

However, several limitations should also be acknowledged. First, this was an observational analysis of pooled trial data, and although we adjusted for an extensive range of prognostic covariates, the association between hemoglobin and outcomes should not be assumed to be causal; also, residual confounding by unmeasured factors,

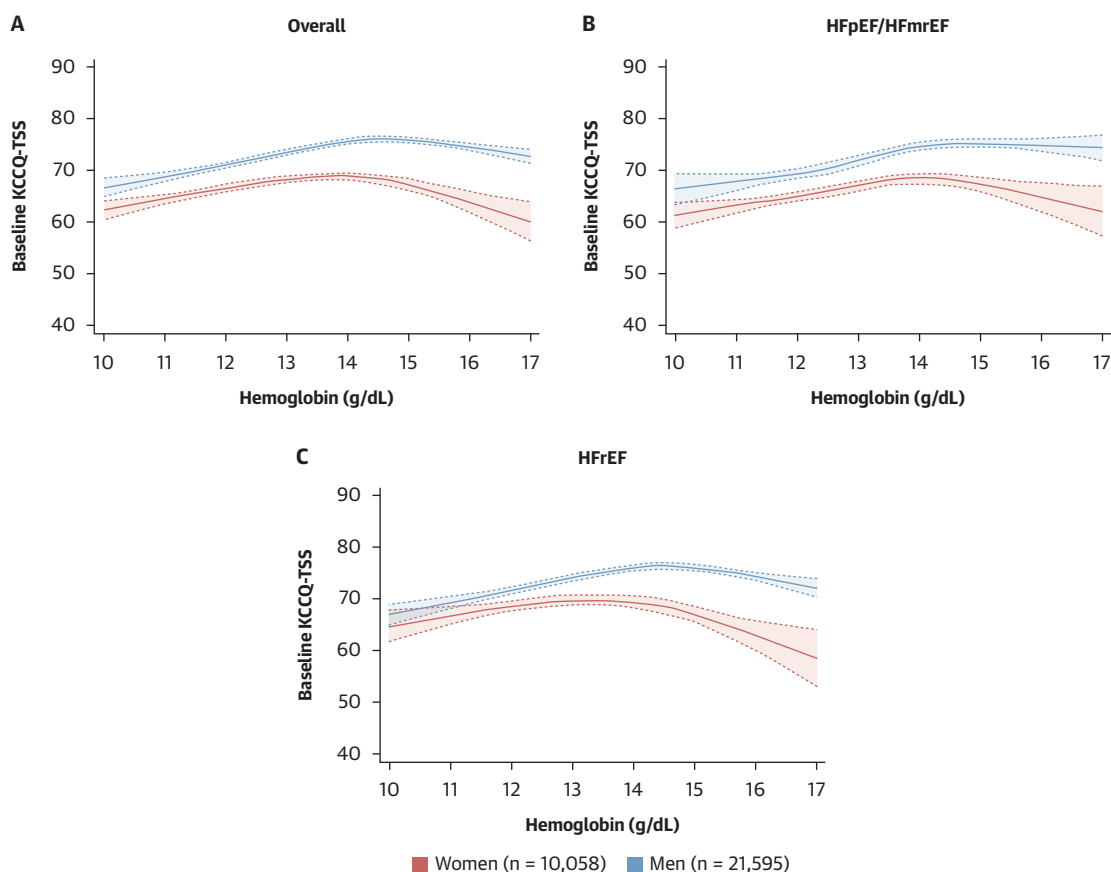
including iron status and inflammatory burden, cannot be excluded. Detailed markers of iron metabolism, such as ferritin and transferrin saturation, were not consistently available across trials and were available at the individual-patient level only in DAPA-HF. The limited sample size and number of events therefore precluded a reliable assessment of whether the prognostic association of hemoglobin was independent of, or modified by, serum iron. Second, hemoglobin was measured once at baseline, and we could not account for changes in hemoglobin over time or for hemoglobin variability, both of

FIGURE 3 Continued



which may independently influence risk. Third, because some trials excluded patients with severe anemia, the lower end of the hemoglobin distribution is likely under-represented relative to unselected HF populations, which may affect the precision of the spline-based estimates at the extremes and limit generalizability. Fourth, the Poisson models used to derive the hemoglobin nadirs and low-risk ranges in the primary analysis were unadjusted, and the derived estimates may therefore have been influenced by differences in clinical characteristics across hemoglobin concentrations. Although adjusted sensitivity analyses using marginal standardization yielded generally similar results, some estimates were less precise, particularly the estimated nadir for

cardiovascular death in women. Finally, because this analysis was restricted to trial participants who met specific eligibility criteria and were generally clinically stable without severe congestion or edema, our findings may not generalize fully to all patients with HF. Although broadly similar associations were observed in a sensitivity analysis restricted to the more recently conducted DAPA-HF, GALACTIC-HF, and FINEARTS-HF trials ([Supplemental Table 17](#)), the use of contemporary guideline-directed medical therapy, including SGLT2 inhibitors, was limited or nonuniform. Therefore, the applicability of our findings to contemporary HF populations receiving comprehensive guideline-directed medical therapy requires further confirmation.

FIGURE 4 KCCQ-TSS Across the Spectrum of Hemoglobin in Patients With Heart Failure Stratified by Sex and Ejection Fraction Phenotype

Overall (A), heart failure with preserved ejection fraction (HFpEF)/heart failure with mildly reduced ejection fraction (HFmrEF) (B), and heart failure with reduced ejection fraction (HFrEF) (C). The x-axis shows hemoglobin concentration, and the y-axis shows the Kansas City Cardiomyopathy Questionnaire total symptom scores (KCCQ-TSS). The solid line shows the spline-estimated KCCQ-TSS, and the dashed lines show the 95% CI.

CONCLUSIONS

This large individual participant-level analysis demonstrates that the hemoglobin concentrations associated with the lowest risk of death and HF hospitalization are consistently higher than the current WHO anemia thresholds in both women and men with HF. This suggests that hemoglobin concentrations above the current WHO anemia thresholds may still be clinically and prognostically relevant in this population. Although these findings should not be interpreted as supporting higher therapeutic hemoglobin targets, they provide outcome-based evidence on the prognostic relevance of lower hemoglobin concentrations in HF and suggest that values just above the current WHO thresholds may prompt consideration of potentially reversible underlying contributors. Prospective studies are needed to

determine whether incorporating hemoglobin-associated risk information into clinical assessment improves risk stratification and facilitates earlier identification of patients who may benefit from targeted investigation and treatment.

DATA AVAILABILITY The original data are available through the specific application process offered by each sponsor of the trials included in the present analyses.

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APPENDIX For supplemental tables and figures, please see the online version of this paper.